

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 04 · Binomial Expansion

With Answers

Questions with full worked solutions

- 43 pages
- Past papers up to January 2026

PURE 2 | TOPIC 04

Binomial Expansion

Your classified route

19

QUESTIONS

124

MARKS

3

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 04 | FAMILY 01

Approximation

WHAT THIS FAMILY ASKS

Recognise the approximation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

1 classified questions

8 total marks

T04

04_APPROX

04_APPROX_CORE

CHAPTER
Binomial Expansion

FAMILY
Approximation

VARIANT
core family move

8 marks

WMA12/01 MAY/JUNE 2021

Question 4

Past-paper question

WMA12-01_2021_MayJune_Q04

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

- (b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

Without calculating 2.0125^{13}

- (c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

T04 Binomial Expansion

04_APPROX Approximation

04_APPROX_CORE core
family move

WORKED SOLUTION

Question 4

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q04

1

Write the binomial terms

$$(2 + \frac{x}{8})^{13} = 2^{13} + \binom{13}{1} 2^{12} \frac{x}{8} + \binom{13}{2} 2^{11} (\frac{x}{8})^2 + \binom{13}{3} 2^{10} (\frac{x}{8})^3 + \dots$$

2

Simplify the expansion

This gives $8192 + 6656x + 2496x^2 + 572x^3 + \dots$.

3

Match the value

Since $2.0125 = 2 + 0.0125 = 2 + \frac{0.1}{8}$, use $x = 0.1$ in the expansion.

4

Calculate the approximation

$$8192 + 6656(0.1) + 2496(0.1)^2 + 572(0.1)^3 = 8883.132.$$

5

Decide the direction

The next terms in the expansion are positive because $x = 0.1 > 0$, so stopping after the x^3 term gives an underestimate.

FINAL ANSWER

WMA12-01_2021_MayJune_Q04

(a) $8192 + 6656x + 2496x^2 + 572x^3$. (b) 8883.132. (c) **Underestimate.**

TOPIC 04 | FAMILY 02

Coefficients

WHAT THIS FAMILY ASKS

Recognise the coefficients demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**8 classified questions**

50 total marks

T04

04_COEFF

04_COEFF_CORE

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Coefficients

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2021

Question 4

Past-paper question

WMA12-01_2021_Jan_Q04

4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

- (b) find the possible values of p .

(4)

T04 Binomial Expansion

04_COEFF Coefficients

04_COEFF_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 January 2021

WMA12-01_2021_Jan_Q04

1

Write the first binomial terms

$$(2 + px)^6 = 2^6 + \binom{6}{1}2^5(px) + \binom{6}{2}2^4(px)^2 + \dots$$

2

Simplify the expansion

$$\text{The first three terms are } 64 + 192px + 240p^2x^2.$$

3

Find the coefficient of x squared

$$\text{In } (3 - \frac{1}{2}x)(2 + px)^6, \text{ the } x^2 \text{ coefficient is } 3(240p^2) - \frac{1}{2}(192p).$$

4

Form the equation

$$\text{So } 720p^2 - 96p = -\frac{3}{4}.$$

5

Solve the quadratic

$$\text{Multiplying by 4 gives } 2880p^2 - 384p + 3 = 0, \text{ or } 960p^2 - 128p + 1 = 0.$$

6

State the values of p

$$p = \frac{128 \pm 112}{1920}, \text{ so } p = \frac{1}{8} \text{ or } p = \frac{1}{120}.$$

FINAL ANSWER

WMA12-01_2021_Jan_Q04

$$(a) 64 + 192px + 240p^2x^2. \quad (b) p = \frac{1}{8} \text{ or } p = \frac{1}{120}.$$

T04

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04_COEFF_CORE

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6 marks

WMA12/01 OCTOBER 2021

Question 1

Past-paper question

WMA12-01_2021_Oct_Q01

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

T04 Binomial Expansion

04_COEFF Coefficients

04_COEFF_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 October 2021

WMA12-01_2021_Oct_Q01

1

Compare the linear termIn $(1 + kx)^{16}$, the coefficient of x is $16k$.

2

Find k Since the coefficient of x is -4 , $16k = -4$, so $k = -\frac{1}{4}$.

3

Find p The coefficient of x^2 is $\binom{16}{2}k^2 = 120\left(\frac{1}{16}\right) = \frac{15}{2}$.

4

Find the needed binomial termsFor $k = -\frac{1}{4}$, the x^2 coefficient in $(1 + kx)^{16}$ is $\frac{15}{2}$ and the x^3 coefficient is $\binom{16}{3}\left(-\frac{1}{4}\right)^3 = -\frac{35}{4}$.

5

Build the x squared term in g In $g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$, the x^2 term comes from $2\left(\frac{15}{2}x^2\right)$ and $\frac{16}{x}\left(-\frac{35}{4}x^3\right)$.

6

SimplifyThe required term is $15x^2 - 140x^2 = -125x^2$.**FINAL ANSWER**

WMA12-01_2021_Oct_Q01

 $(a)(i) k = -\frac{1}{4}$. $(a)(ii) p = \frac{15}{2}$. $(b) -125x^2$.

T04

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7 marks

WMA12/01 JANUARY 2022

Question 3

Past-paper question

WMA12-01_2022_Jan_Q03

3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of $f(x)$, the constant term is 3 times the coefficient of x .

- (b) Find the value of k .

(3)

T04 Binomial Expansion

04_COEFF Coefficients

04_COEFF_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 January 2022

WMA12-01_2022_Jan_Q03

1

Write the first terms

$$\left(2 - \frac{kx}{4}\right)^8 = 2^8 + \binom{8}{1}2^7\left(-\frac{kx}{4}\right) + \binom{8}{2}2^6\left(-\frac{kx}{4}\right)^2 + \binom{8}{3}2^5\left(-\frac{kx}{4}\right)^3 + \dots$$

2

Simplify the expansion

The first four terms are $256 - 256kx + 112k^2x^2 - 28k^3x^3$.

3

Find the constant term in f

For $f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$, the constant term is $5(256) = 1280$.

4

Find the coefficient of x

The coefficient of x is $5(-256k) - 3(256) = -1280k - 768$.

5

Use the condition

The constant term is 3 times the coefficient of x , so $1280 = 3(-1280k - 768)$.

6

Solve for k

$$1280 = -3840k - 2304, \text{ hence } k = -\frac{3584}{3840} = -\frac{14}{15}.$$

FINAL ANSWER

WMA12-01_2022_Jan_Q03

$$(a) 256 - 256kx + 112k^2x^2 - 28k^3x^3. \quad (b) k = -\frac{14}{15}.$$

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WMA12/01 OCTOBER 2022

Question 2

Past-paper question

WMA12-01_2022_Oct_Q02

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

- (a) show that $k = \frac{2}{5}$

(2)

- (b) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

- (c) find the gradient of C at the point where $x = 0$

(2)

T04 Binomial Expansion

04_COEFF Coefficients

04_COEFF_CORE core family
move

WORKED SOLUTION

Question 2

WMA12/01 October 2022

WMA12-01_2022_Oct_Q02

1

Use the remainder theoremDividing by $4x - 5$ means the remainder is $f\left(\frac{5}{4}\right)$.

2

Substitute x equals five quarters

$$f\left(\frac{5}{4}\right) = \left(2 - \frac{5k}{4}\right)^5 = \frac{243}{32}.$$

3

Take the fifth rootSince $\frac{243}{32} = \left(\frac{3}{2}\right)^5$, we have $2 - \frac{5k}{4} = \frac{3}{2}$.

4

Find k

$$\frac{5k}{4} = \frac{1}{2}, \text{ so } k = \frac{2}{5}.$$

5

Expand the binomial

$$\text{With } k = \frac{2}{5}, f(x) = \left(2 - \frac{2}{5}x\right)^5.$$

6

Find the first three terms

$$2^5 + 5(2^4)\left(-\frac{2}{5}x\right) + 10(2^3)\left(-\frac{2}{5}x\right)^2 = 32 - 32x + \frac{64}{5}x^2.$$

7

Use the coefficient of x

For a polynomial, the gradient at $x = 0$ is the coefficient of x in the expansion. Therefore $f'(0) = -32$.

FINAL ANSWER

WMA12-01_2022_Oct_Q02

$$(a) k = \frac{2}{5}. \quad (b) 32 - 32x + \frac{64}{5}x^2. \quad (c) -32.$$

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WMA12/01 MAY/JUNE 2023

Question 4

Past-paper question

WMA12-01_2023_MayJune_Q04

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A , B , C and D are constants.

- (a) Find the value of A

(1)

Given that

- $B = 18D$
- $p < 0$

- (b) find

- (i) the value of p
- (ii) the value of C

(6)

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WORKED SOLUTION

Question 4

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q04

1

Find the constant termIn $(3 + px)^5$, the constant term is $3^5 = 243$, so $A = 243$.

2

Find BThe coefficient of x is $5(3^4)p = 405p$, so $B = 405p$.

3

Find DThe coefficient of x^3 is $\binom{5}{3}(3^2)p^3 = 10 \cdot 9p^3 = 90p^3$, so $D = 90p^3$.

4

Use B equals 18D $405p = 18(90p^3) = 1620p^3$.

5

Solve for pSince $p < 0$, $p \neq 0$. Divide by $405p$ to get $1 = 4p^2$, so $p = \pm \frac{1}{2}$. Therefore $p = -\frac{1}{2}$.

6

Find CThe coefficient of x^2 is $\binom{5}{2}(3^3)p^2 = 10 \cdot 27 \cdot \frac{1}{4} = \frac{135}{2}$.**FINAL ANSWER**

WMA12-01_2023_MayJune_Q04

(a) $A = 243$. (b)(i) $p = -\frac{1}{2}$. (b)(ii) $C = \frac{135}{2}$.

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3 marks

WMA12/01 JANUARY 2024

Question 2

Past-paper question

WMA12-01_2024_Jan_Q02

2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

T04 Binomial Expansion

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move

WORKED SOLUTION

Question 2

WMA12/01 January 2024

WMA12-01_2024_Jan_Q02

1

Choose the x power

In $\left(\frac{3}{8} + 4x\right)^{12}$, the term in x^7 comes from choosing $(4x)^7$ and $\left(\frac{3}{8}\right)^5$.

2

Write the term

The coefficient is $\binom{12}{7} \left(\frac{3}{8}\right)^5 4^7$.

3

Simplify the powers

$4^7 = 2^{14}$ and $8^5 = 2^{15}$, so $\left(\frac{3}{8}\right)^5 4^7 = \frac{3^5}{2}$.

4

Evaluate

$\binom{12}{7} = 792$, so the coefficient is $792 \cdot \frac{243}{2} = 96228$.

FINAL ANSWER

96228.

WMA12-01_2024_Jan_Q02

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6 marks

WMA12/01 OCTOBER 2025

Question 2

Past-paper question

WMA12-01_2025_Oct_Q02

2. The first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + px)^{10}$$

are

$$1 + 15x + qx^2 + rx^3$$

where p , q and r are constants.

Find the value of p , the value of q and the value of r .

(6)

T04 Binomial Expansion

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WORKED SOLUTION

Question 2

WMA12/01 October 2025

WMA12-01_2025_Oct_Q02

1

Write the first four terms

$$(1 + px)^{10} = 1 + 10px + \binom{10}{2}p^2x^2 + \binom{10}{3}p^3x^3 + \dots$$

2

Compare the x coefficient

Since the coefficient of x is 15, $10p = 15$, so $p = \frac{3}{2}$.

3

Find q

$$q = \binom{10}{2}p^2 = 45 \left(\frac{3}{2}\right)^2 = \frac{405}{4}.$$

4

Find r

$$r = \binom{10}{3}p^3 = 120 \left(\frac{3}{2}\right)^3 = 405.$$

FINAL ANSWER

$$p = \frac{3}{2}, q = \frac{405}{4}, r = 405.$$

WMA12-01_2025_Oct_Q02

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6 marks

WMA12/01 JANUARY 2026

Question 1

Past-paper question

WMA12-01_2026_Jan_Q01

1. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + kx)^7$$

where k is a non-zero constant. Give each term in simplest form.

(4)

Given that, in this expansion, the coefficient of x^2 is 3 times the coefficient of x^3

- (b) find, in simplest form, the value of k .

(2)

T04 Binomial Expansion

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WORKED SOLUTION

Question 1

WMA12/01 January 2026

WMA12-01_2026_Jan_Q01

1

Use the binomial expansionFor $(3 + kx)^7$, the first four terms are $3^7 + \binom{7}{1}3^6(kx) + \binom{7}{2}3^5(kx)^2 + \binom{7}{3}3^4(kx)^3$.

2

Simplify the coefficientsThis gives $2187 + 5103kx + 5103k^2x^2 + 2835k^3x^3$.

3

Compare the required coefficientsThe coefficient of x^2 is $5103k^2$ and the coefficient of x^3 is $2835k^3$.

4

Solve for kSince $5103k^2 = 3(2835k^3)$ and $k \neq 0$, divide by k^2 to get $5103 = 8505k$, hence $k = \frac{5103}{8505} = \frac{3}{5}$.**FINAL ANSWER**

WMA12-01_2026_Jan_Q01

(a) $2187 + 5103kx + 5103k^2x^2 + 2835k^3x^3$. (b) $k = \frac{3}{5}$.

TOPIC 04 | FAMILY 03

Expansion Terms

WHAT THIS FAMILY ASKS

Recognise the expansion terms demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

10 classified questions

66 total marks

T04

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6 marks

WMA12/01 OCTOBER 2019

Question 3

Past-paper question

WMA12-01_2019_Oct_Q03

3. (a) Find the first 4 terms, in ascending powers of x , in the binomial expansion of

$$\left(1 + \frac{x}{4}\right)^{12}$$

giving each coefficient in its simplest form.

(3)

- (b) Find the term independent of x in the expansion of

$$\left(\frac{x^2 + 8}{x^5}\right)\left(1 + \frac{x}{4}\right)^{12}$$

(3)

T04 Binomial Expansion

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WORKED SOLUTION

Question 3

WMA12/01 October 2019

WMA12-01_2019_Oct_Q03

1

Write the binomial terms

$$(1 + \frac{x}{4})^{12} = 1 + \binom{12}{1} \frac{x}{4} + \binom{12}{2} (\frac{x}{4})^2 + \binom{12}{3} (\frac{x}{4})^3 + \dots$$

2

Simplify the first four terms

$$\text{This gives } 1 + 3x + \frac{33}{8}x^2 + \frac{55}{16}x^3.$$

3

Rewrite the prefactor

$$\frac{x^2+8}{x^5} = x^{-3} + 8x^{-5}.$$

4

Find the constant contributions

The x^{-3} part needs the x^3 coefficient, which is $\frac{55}{16}$. The $8x^{-5}$ part needs the x^5 coefficient, which is $8\binom{12}{5}(\frac{1}{4})^5 = 8 \cdot \frac{99}{128} = \frac{99}{16}$.

5

Add the constant terms

$$\text{The term independent of } x \text{ is } \frac{55}{16} + \frac{99}{16} = \frac{154}{16} = \frac{77}{8}.$$

FINAL ANSWER

WMA12-01_2019_Oct_Q03

$$(a) 1 + 3x + \frac{33}{8}x^2 + \frac{55}{16}x^3. \quad (b) \frac{77}{8}.$$

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7 marks

WMA12/01 JANUARY 2020

Question 2

Past-paper question

WMA12-01_2020_Jan_Q02

2. One of the terms in the binomial expansion of $(3 + ax)^6$, where a is a constant, is $540x^4$

(a) Find the possible values of a .

(4)

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6$$

(3)

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WORKED SOLUTION

Question 2

WMA12/01 January 2020

WMA12-01_2020_Jan_Q02

1

Identify the x to the fourth termIn $(3 + ax)^6$, the x^4 term is $\binom{6}{4}3^2(ax)^4$.

2

Form the equation for a $\binom{6}{4}3^2a^4x^4 = 15 \cdot 9a^4x^4 = 135a^4x^4$. Since this term is $540x^4$, $135a^4 = 540$.

3

Solve for a $a^4 = 4$, so the real possible values are $a = \sqrt{2}$ or $a = -\sqrt{2}$.

4

Find the constant contribution from 1 over 81The constant term from $\frac{1}{81}(3 + ax)^6$ is $\frac{1}{81} \cdot 3^6 = 9$.

5

Find the constant contribution from x to the minus sixThe x^6 term in $(3 + ax)^6$ is a^6x^6 . Since $a = \pm\sqrt{2}$, $a^6 = 8$, so $\frac{1}{x^6} \cdot a^6x^6$ contributes 8.

6

Add the constant termsThe term independent of x is $9 + 8 = 17$.**FINAL ANSWER**(a) $a = \pm\sqrt{2}$. (b) 17.

WMA12-01_2020_Jan_Q02

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4 marks

WMA12/01 MAY/JUNE 2022

Question 1

Past-paper question

WMA12-01_2022_MayJune_Q01

1. Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(4)

T04 Binomial Expansion

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WORKED SOLUTION

Question 1

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q01

1

Write the binomial terms

$$(2 + \frac{3}{8}x)^{10} = 2^{10} + \binom{10}{1}2^9\left(\frac{3}{8}x\right) + \binom{10}{2}2^8\left(\frac{3}{8}x\right)^2 + \binom{10}{3}2^7\left(\frac{3}{8}x\right)^3 + \dots$$

2

Simplify the constant and linear terms

$$2^{10} = 1024 \text{ and } \binom{10}{1}2^9\left(\frac{3}{8}\right)x = 1920x.$$

3

Simplify the quadratic term

$$\binom{10}{2}2^8\left(\frac{3}{8}\right)^2x^2 = 1620x^2.$$

4

Simplify the cubic term

$$\binom{10}{3}2^7\left(\frac{3}{8}\right)^3x^3 = 810x^3.$$

5

State the expansion

$$\text{The first four terms are } 1024 + 1920x + 1620x^2 + 810x^3.$$

FINAL ANSWER

WMA12-01_2022_MayJune_Q01

$$1024 + 1920x + 1620x^2 + 810x^3.$$

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WMA12/01 JANUARY 2023
Question 3

Past-paper question

WMA12-01_2023_Jan_Q03

3. $f(x) = \left(2 + \frac{kx}{8}\right)^7$ where k is a non-zero constant

- (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $f(x)$.
Give each term in simplest form.

(4)

Given that, in the binomial expansion of $f(x)$, the coefficients of x , x^2 and x^3 are the first 3 terms of an arithmetic progression,

- (b) find, using algebra, the possible values of k .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

T04 Binomial Expansion

04_TERM Expansion terms

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move

WORKED SOLUTION

Question 3

WMA12/01 January 2023

WMA12-01_2023_Jan_Q03

1

Start the binomial expansion

$(2 + \frac{kx}{8})^7$ begins with $2^7 + 7(2^6)(\frac{kx}{8}) + (\frac{7}{2})(2^5)(\frac{kx}{8})^2 + (\frac{7}{3})(2^4)(\frac{kx}{8})^3$.

2

Simplify the terms

This gives $128 + 56kx + \frac{21}{2}k^2x^2 + \frac{35}{32}k^3x^3$.

3

Use the arithmetic progression condition

The coefficients of x , x^2 and x^3 are $56k$, $\frac{21}{2}k^2$ and $\frac{35}{32}k^3$.

4

Set middle twice

For three terms in an arithmetic progression, twice the middle term equals the sum of the outer terms: $21k^2 = 56k + \frac{35}{32}k^3$.

5

Use k non-zero

Since $k \neq 0$, divide by k and multiply by 32: $672k = 1792 + 35k^2$.

6

Solve the quadratic

So $35k^2 - 672k + 1792 = 0$. The discriminant is $672^2 - 4(35)(1792) = 200704 = 448^2$.

7

Find the possible values

$k = \frac{672 \pm 448}{70}$, giving $k = 16$ or $k = \frac{16}{5}$.

FINAL ANSWER

WMA12-01_2023_Jan_Q03

(a) $128 + 56kx + \frac{21}{2}k^2x^2 + \frac{35}{32}k^3x^3$. (b) $k = \frac{16}{5}$ or $k = 16$.

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WMA12/01 OCTOBER 2023

Question 10

Past-paper question

WMA12-01_2023_Oct_Q10

10. (i) (a) Find, in ascending powers of x , the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6 \quad (3)$$

For a particular value of x , these three terms form consecutive terms in a geometric series.

- (b) Find this value of x . (3)

- (ii) In a **different** geometric series,

- the first term is $\sin^2 \theta$
- the common ratio is $2 \cos \theta$
- the sum to infinity is $\frac{8}{5}$

- (a) Show that

$$5 \cos^2 \theta - 16 \cos \theta + 3 = 0 \quad (3)$$

- (b) Hence find the exact value of the 2nd term in the series. (3)

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WORKED SOLUTION

Question 10

WMA12/01 October 2023

WMA12-01_2023_Oct_Q10

1

Find the second termThe 2nd term of $(3 + 2x)^6$ is $\binom{6}{1}3^5(2x) = 6 \cdot 243 \cdot 2x = 2916x$.

2

Find the third termThe 3rd term is $\binom{6}{2}3^4(2x)^2 = 15 \cdot 81 \cdot 4x^2 = 4860x^2$.

3

Find the fifth termThe 5th term is $\binom{6}{4}3^2(2x)^4 = 15 \cdot 9 \cdot 16x^4 = 2160x^4$.

4

Use the geometric condition

For three consecutive terms in a geometric series, the middle term squared equals the product of the two outer terms.

5

Set up the equation

$$(4860x^2)^2 = (2916x)(2160x^4).$$

6

Solve for xFor the non-zero geometric case, divide by x^4 : $4860^2 = 2916 \cdot 2160x$, so $x = \frac{4860^2}{2916 \cdot 2160} = \frac{15}{4}$.

7

Use the sum to infinityIn the different geometric series, first term $= \sin^2 \theta$, common ratio $= 2 \cos \theta$, and $S_\infty = \frac{8}{5}$.

8

Form the equation

$$\frac{\sin^2 \theta}{1 - 2 \cos \theta} = \frac{8}{5}.$$

9

Use sine squared $5 \sin^2 \theta = 8(1 - 2 \cos \theta)$. With $\sin^2 \theta = 1 - \cos^2 \theta$, this becomes $5(1 - \cos^2 \theta) = 8 - 16 \cos \theta$.

10

Show the required equationRearranging gives $5 \cos^2 \theta - 16 \cos \theta + 3 = 0$.

11

Solve for cosine $5 \cos^2 \theta - 16 \cos \theta + 3 = 0$ gives $\cos \theta = 3$ or $\cos \theta = \frac{1}{5}$. The possible value is $\cos \theta = \frac{1}{5}$.

12

Find the second termThe 2nd term is first term times common ratio: $\sin^2 \theta (2 \cos \theta) = \left(1 - \frac{1}{25}\right) \cdot \frac{2}{5} = \frac{24}{25} \cdot \frac{2}{5} = \frac{48}{125}$.**FINAL ANSWER**

WMA12-01_2023_Oct_Q10

(i)(a) $2916x, 4860x^2, 2160x^4$. (i)(b) $x = \frac{15}{4}$. (ii)(a) **Proven.** (ii)(b) $\frac{48}{125}$.

DR ESLAM AHMED

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5 marks

WMA12/01 MAY/JUNE 2024

Question 1

Past-paper question

WMA12-01_2024_MayJune_Q01

1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(1 - \frac{1}{6}x\right)^9$$

giving each term in simplest form.

(3)

- (b) Hence find the coefficient of x^3 in the expansion of

$$(10x + 3)\left(1 - \frac{1}{6}x\right)^9$$

giving the answer in simplest form.

(2)

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WORKED SOLUTION

Question 1

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q01

1

Expand the first terms

$$\left(1 - \frac{1}{6}x\right)^9 = 1 + 9\left(-\frac{1}{6}x\right) + \binom{9}{2}\left(-\frac{1}{6}x\right)^2 + \binom{9}{3}\left(-\frac{1}{6}x\right)^3 + \dots$$

2

Simplify

The first four terms are $1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$.

3

Identify contributions to x cubed

In $(10x + 3)\left(1 - \frac{1}{6}x\right)^9$, the x^3 coefficient comes from 3 times the x^3 coefficient and $10x$ times the x^2 coefficient.

4

Calculate the coefficient

$$3\left(-\frac{7}{18}\right) + 10(1) = -\frac{7}{6} + 10 = \frac{53}{6}.$$

FINAL ANSWER

WMA12-01_2024_MayJune_Q01

$$(a) 1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3. \quad (b) \frac{53}{6}.$$

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WMA12/01 MAY/JUNE (R) 2024

Question 4

Past-paper question

WMA12-01_2024_MayJuneR_Q04

4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

- (b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

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WORKED SOLUTION

Question 4

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q04

1

Write the expansion terms

$$(3 + 2x)^6 = 3^6 + \binom{6}{1}3^5(2x) + \binom{6}{2}3^4(2x)^2 + \binom{6}{3}3^3(2x)^3 + \dots$$

2

Simplify the first four termsThis gives $729 + 2916x + 4860x^2 + 4320x^3$.

3

Identify x squared contributionsIn $(2x^2 - \frac{1}{6x})(3 + 2x)^6$, the x^2 term comes from $2x^2$ times the constant term and $-\frac{1}{6x}$ times the x^3 term.

4

Calculate the coefficientThe coefficient is $2(729) - \frac{1}{6}(4320) = 1458 - 720 = 738$.**FINAL ANSWER**

WMA12-01_2024_MayJuneR_Q04

(a) $729 + 2916x + 4860x^2 + 4320x^3$. (b) 738.

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WMA12/01 OCTOBER 2024

Question 5

Past-paper question

WMA12-01_2024_Oct_Q05

5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

- (b) find the value of a .

(4)

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WORKED SOLUTION

Question 5

WMA12/01 October 2024

WMA12-01_2024_Oct_Q05

1

Write the first three binomial terms

$$(2 + ax)^6 = 2^6 + 6(2^5)(ax) + \binom{6}{2}(2^4)(ax)^2 + \dots$$

2

Simplify the termsThe first three terms are $64 + 192ax + 240a^2x^2$.

3

Expand the other factor

$$\left(3 + \frac{1}{x}\right)^2 = 9 + 6x^{-1} + x^{-2}.$$

4

Collect constant contributionsThe constant term in $\left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$ is $9(64) + 6(192a) + 240a^2$.

5

Use the given constant term

$$576 + 1152a + 240a^2 = 576, \text{ so } 240a^2 + 1152a = 0.$$

6

Solve for a

$$48a(5a + 24) = 0. \text{ Since } a \neq 0, 5a + 24 = 0, \text{ so } a = -\frac{24}{5}.$$

FINAL ANSWER

WMA12-01_2024_Oct_Q05

$$(a) 64 + 192ax + 240a^2x^2. \quad (b) a = -\frac{24}{5}.$$

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WMA12/01 JANUARY 2025

Question 2

Past-paper question

WMA12-01_2025_Jan_Q02

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(2 - 5x)^8$$

giving each term in simplest form.

(4)

This expansion is to be used to find an approximation for 2.05^8

- (b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

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WORKED SOLUTION

Question 2

WMA12/01 January 2025

WMA12-01_2025_Jan_Q02

1

Write the first four terms

$$(2 - 5x)^8 = 2^8 + 8(2^7)(-5x) + \binom{8}{2}(2^6)(-5x)^2 + \binom{8}{3}(2^5)(-5x)^3 + \dots$$

2

Simplify each term

The first four terms are $256 - 5120x + 44800x^2 - 224000x^3$.

3

Connect the expansion to 2.05

To use $(2 - 5x)^8$ for 2.05^8 , set $2 - 5x = 2.05$.

4

Find x

$$-5x = 0.05, \text{ so } x = -0.01.$$

FINAL ANSWER

WMA12-01_2025_Jan_Q02

$$(a) 256 - 5120x + 44800x^2 - 224000x^3. \quad (b) x = -0.01.$$

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6 marks

WMA12/01 MAY/JUNE 2025

Question 1

Past-paper question

WMA12-01_2025_MayJune_Q01

- 1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

In the series expansion of

$$(5 + kx)(1 - 4x)^7$$

where k is a constant

the coefficient of the term in x^2 is 1316

- (b) Use the answer to part (a) to find the value of k .

(2)

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WORKED SOLUTION

Question 1

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q01

1

Write the binomial terms

$$(1 - 4x)^7 = 1 + 7(-4x) + \binom{7}{2}(-4x)^2 + \binom{7}{3}(-4x)^3 + \dots$$

2

Simplify

The first four terms are $1 - 28x + 336x^2 - 2240x^3$.

3

Find the x squared terms

In $(5 + kx)(1 - 4x)^7$, the x^2 coefficient comes from $5(336x^2)$ and $(kx)(-28x)$.

4

Use the given coefficient

$$5(336) - 28k = 1316, \text{ so } 1680 - 28k = 1316.$$

5

Find k

$$28k = 364, \text{ hence } k = 13.$$

FINAL ANSWER

WMA12-01_2025_MayJune_Q01

$$(a) 1 - 28x + 336x^2 - 2240x^3. \quad (b) k = 13.$$